Early Detection of Lacunar Stroke for Early Diagnosis Using an AI-Powered System for Continuous Bilateral Sensory Asymmetry Monitoring

Name: Morane Adisayam Angemia
ID: 2403_27680

Name: Diya Devi Brijrajh
ID: 2403_27694

Name: Loïc Fred Cheerkoot
ID: 2403_27696

Name: Bibi Ra'eesah Goolam Ossen
ID: 2403_27707

***Abstract*—Lacunar Infarct (LI) is an underdiagnosed subtype of ischemic stroke due to its subtle symptoms and bilateral sensory asymmetry that are often not detected through current monitoring systems. Thus, early identification of these deficits is critical for timely intervention to prevent progressive neurological damage. This study aimed to develop a simulation-based AI-powered system for the continuous monitoring of bilateral sensory asymmetry to enable early detection of lacunar stroke. Bilateral sensory asymmetry was analyzed using a three-tier system that included a Random Forest model, a MariaDB database, and a Flask-based web interface. To determine the patterns of stuttering LI, volatility and asymmetry characteristics were computed on 5,000 simulated patient profiles. The random forest classifier produced 100% accuracy on all five tiers of severity, and the volatility characteristics were found to dominate the predictive power at 37%. The model produced perfect discriminative ability (AUC 0.98) on all levels of impact, which validates that the temporal sensory analysis can capture the dynamic pathophysiology of active small vessel disease. This paper proposes the first AI-powered system for continuous bilateral sensory asymmetry monitoring specifically designed for lacunar stroke. This system provides a proactive, clinically actionable approach to early stroke prevention by detecting pre-symptomatic risk through real-time analysis of stuttering sensory patterns.**

Keywords*—Lacunar infarct, ischemic stroke, sensory asymmetry, continuous monitoring, early detection, machine learning, clinical decision support tool.

I. Introduction

The Lacunar Infarct (LI) is one of the most prevalent yet often misidentified stroke subtypes. It is caused by the blockage of tiny, deep perforating arteries in the subcortical regions of the brain and accounts for 20%-35% of all ischemic strokes [1]. Subtle sensory loss, such as numbness, is an early warning sign for LI. This symptom is usually temporary and unilateral but is often dismissed by patients or even healthcare practitioners [2]. Failure to detect these early signs of latent sensory asymmetry delays diagnosis and treatment, even though it is a crucial indicator of a potential stroke. Current clinical practice relies heavily on subjective self-reports and limited clinical assessments, which can lead to neurological disorders or functional impairment [3,4]. Therefore, there is an essential requirement for a reliable, continuous, real-time monitoring and fully automated system to detect these early subtle sensory asymmetries and enable timely intervention [5, 6]. Patients with LI typically present with a classical lacunar syndrome [8], which includes pure motor hemiparesis, pure sensory syndrome, sensorimotor stroke, ataxic hemiparesis, or dysarthria-clumsy hand. The major risk factors for lacunar stroke include Diabetes Mellitus and Hypertension [8]. LI exhibits a paradoxical clinical history knowing it has a good short-term prognosis marked by low early mortality and decreased functional disability upon hospital discharge, but carries an elevated mid- and long-term risk of death, stroke recurrence, and dementia [8]. A common characteristic of LI is the asymptomatic progression of small-vessel disease [8]. For this reason, LI should be recognized as a potentially serious neurological condition rather than a benign disorder, necessitating comprehensive management strategies and vigilant long-term follow-up for affected patients [8].

To address the challenge of stroke detection, significant research has focused on AI-powered wearable and mobile technologies. Wearable devices, such as smartwatches and biosensors, have been developed to monitor key physiological indicators like blood pressure, heart rhythm, and autonomic nervous system activity [9]. These devices provide continuous, real-time data streams that feed into machine-learning models to assess general cardiovascular and stroke risk [5, 9]. While valuable for identifying systemic risk factors, these systems are not designed to detect the specific, focal neurological deficits of an evolving Lacunar Infarct. Concurrently, research has explored using smartphones for direct symptom assessment. The FAST.AI application is a prominent example as it incorporates a smartphone's camera, motion sensor, and microphone with machine learning algorithms to identify signs of acute stroke like facial asymmetry and arm weakness [10]. When tested on acute stroke patients, it demonstrated high accuracy for these overt motor symptoms [10]. Another approach uses a smartphone's videophone function to record and analyze specific facial movements, providing a high-accuracy method for detecting facial nerve palsy, a sign that

may be associated with stroke risk [11]. However, these applications target visible, gross motor impairments such as significant weakness or facial droop which is a characteristic of large-vessel strokes. They are not suited to capture the subtle, purely sensory, or pre-symptomatic changes that are the hallmark early warnings of LI [10].

Current stroke detection approaches including systems like FAST.AI model, are unable to capture the silent infarct formation or microvascular alterations and fail to detect the subtle diagnoses like Lacunar Infarcts (LI) [10]. Besides that there are no mobile applications that include continuous AI-automated monitoring for LI syndromes which in result makes early intervention more difficult and often misses the preventive measures. The conventional methods for Lateral Ischemia (LI) diagnosis using clinical and imaging methods are insufficient for continuous monitoring as they are episodic, costly and dependent on patient self-reporting [20]. Additionally, instruments like the NIH Stroke Scale (NIHSS) lack the sensitivity to detect subtle abnormalities associated with LI, such as minor brain infarcts or transient symptoms, demonstrating the limitations of non-technical approaches in the early detection of LI [21]. By creating an AI-powered IT solution, this study promotes an initiative toward proactively screening for Lacunar Infarct (LI) risk. It shifts the focus from common stroke symptoms to identifying the distinct pathophysiological indicators of LI, including subtle, pre-symptomatic, and frequently bilateral asymmetry. This approach requires highly sophisticated computational techniques to guarantee continuous, unbiased LI-specific monitoring [5]. The work focuses on developing a software tool for early identification of lacunar infarctions (LI). This tool employs a machine learning model trained on structured clinical datasets to identify patterns in early symptoms. It assesses both objective and subjective clinical data like affected anatomical locations, sensation nature and intensity, and vascular risk factors to calculate a risk assessment score for lacunar infarction. The ultimate goal is to create a clinical decision support tool that assesses early symptoms and identifies high-risk cases for immediate evaluation [22].

II. LITERATURE REVIEW

A. *Concept of Lacunar Infarct and its Risk Factors*

Lacunar Infarct (LI) is a small subcortical infarct, which is usually a diameter of 3-15mm in the chronic stage and can be up to 20mm in the acute stage, thus representing infarction in the territory of a single perforating arteriole which affects brain components like the basal ganglia, thalamus, internal capsule, and pons [23, 24]. LI is a subtype of ischemic stroke, representing a typical sign of cerebral small vessel disease (cSVD) [23]. Lipohyalinosis, being the most predominant vascular pathology, is a microangiopathic process that is characterized by segmental arteriolar disorganization, fibrinoid necrosis, and luminal occlusion [23]. The most significant modifiable risk factor is hypertension, which leads to endothelial dysfunction, poor cerebral autoregulation, and blood-brain barrier disruption, hence eventually contributing to arteriolar remodeling and infarction [23]. Through mechanisms such as inflammatory endothelial damage, oxidative stress, and glycation, diabetes mellitus also contributes to microvascular injury. Moreover, risk factors include aging, hypercholesterolaemia, genetic tendency including monogenic disorders such as CADASIL and CO41A1-related cSVD, and smoking habits [23]. Hence, chronic exposure to these risk factors is the pathological sequence that gradually leads to progressive arteriolar constriction, focal ischemia, and eventually infarction [23, 25]. Initially, many lacunar infarcts silent and are detected accidentally on neuroimaging due their small size, chronic ischemic preconditioning and location in the non-strategic brain regions such as the corona radiata instead of the eloquent areas such as the internal capsule or the brainstem, where lacunar infarcts can produce clinically significant deficits [26]. The association of silent infarcts with a higher risk of vascular cognitive impairment, recurrent stroke, and progressive white matter remains clinically significant [23, 24]. As silent infarcts are considered as the first warning indicator of underlying vascular damage, it emphasizes the importance of early detection as the brain damage worsens over time, it can even accelerate cognitive decline and increase the risk of symptomatic stroke if not intervened early for treatment.

B. *Clinical Manifestation and Stuttering Patterns*

The early clinical presentation of lacunar stroke is defined by a subtle typology of sensory changes and motor fluctuations, where bilateral yet asymmetric patterns often serve as critical but frequently overlooked precursors to acute infarction. A landmark clinical study of 99 patients identifies pure sensory stroke as a distinct entity characterized by purely subjective sensory disturbances without motor involvement, differing significantly from pure motor or sensorimotor lacunas in both its presentation and generally more favorable short-term prognosis [2]. These sensory presentations manifest through a spectrum of paresthesia (tingling), numbness, and distinct thermal perception differences, alongside neurogenic pain which often appears as atypical burning or aching sensations [27]. Motor symptoms are often characterized by a specific prodromal phenomenon known as capsular warning syndrome, involving episodes of fluctuation, one-sided weakness [30]. Beyond raw strength deficits, these motor manifestations frequently include coordination difficulties and limb clumsiness, which are hallmark features of ataxic hemiparesis, a syndrome where weakness is accompanied by significant motor incoordination [32].

A critical distinction exists between transient symptoms with lacunar infarction (TSI), which often present as prodromal sensory shifts, and the persistent, lasting symptoms that remain after the acute event [31]. Objective testing utilizing the Rivermead Assessment of Somatosensory Performance (RASP) has demonstrated that deficits are frequently bilateral but asymmetric, even in syndromes traditionally classified as focal events. This rigorous methodology involves bilateral testing across 10 specific regions, including the face (cheek), shoulder, palm, leg, and dorsum of the foot, across seven distinct modalities, which are surface pressure, surface localization, pinprick, proprioception, two-point discrimination, stereognosis, and temperature discrimination. Among these, hot-cold temperature discrimination is one of the most sensitive indicators, as it evaluates the patient's ability to

differentiate thermal stimuli; impairment in this specific modality often signals damage to the spinothalamic tracts before gross touch sensation is lost [29].

Research has established that lacunar infarcts frequently manifest with a characteristic stuttering pattern. The stuttering pattern, also known as the Stuttering Lacunar Syndrome (SLS) is characterized as fluctuating neurological deficits that worsens and improve over short periods, is a well-documented feature of acute lacunar events. This instability reflects the dynamic pathophysiology of small-vessel occlusions, where collateral circulation attempts and penumbral changes create a clinical picture of "in-and-out" symptom variation. These fluctuations can occur over minutes to hours, creating a distinct volatility signature that contrasts with the stable deficits typically seen in large vessel occlusions [33]. Lacunar strokes also often demonstrate a gradual onset pattern, where neurological deterioration evolves progressively over hours or even days rather than reaching maximal severity instantaneously [34]. This temporal evolution reflects the slow expansion of ischemic injury within the deep brain structures supplied by perforating arteries. The gradual progression may involve stepwise worsening or a slowly progressive decline in motor or sensory function, which can be quantified as a negative velocity in neurological scoring systems.

Despite these measurable markers, a significant gap in patient awareness persists. The failure to seek immediate care is often driven by a fundamental lack of awareness concerning stroke symptoms, particularly among older populations who may not recognize early warning signs. While some symptoms like numbness are recognized by only 41% of the population, others such as dizziness (vertigo) and poor visibility frequently fail to be categorized as stroke-related by the majority of the general public [28]. Ultimately, the nuanced nature of these symptoms, from prodromal TSI and thermal discrimination failures to the differentiation between pure sensory and sensorimotor syndromes, underscores the necessity for objective, continuous bilateral monitoring to bridge the gap between clinical reality and subjective patient interpretation.

C. *Diagnostic Challenges and Monitoring Gaps*

Despite medical progress in stroke recognition, the early detection of lacunar infarcts has a poor early diagnostic yield due to technological as well as diagnostic limitations. The National Institutes of Health Stroke Scale (NIHSS) is recognized as one of the main standardized neurological assessment tools for stroke; it is commonly used but demonstrates reduced sensitivity to lacunar infarcts, especially during the pre-symptomatic or minimally symptomatic phase. Lacunar strokes are often associated with subtle motor or sensory impairments that result in poor or zero NIHSS scores, resulting in delayed detection [35]. Furthermore, neurological tests are carried out irregularly during scheduled healthcare appointments, while the evolution of lacunar infarction progresses over time. Neuroimaging, while acting as a key to diagnosis, is also accompanied by its set of drawbacks. Magnetic resonance imaging (MRI) is identified as the gold standard for identifying lacunar infarcts, precisely diffusion-weighted imaging; though it is costly, constrained access and infrastructure requirements result in impracticality for continuous or repeated monitoring [36, 37]. Although the availability of Computed Tomography (CT) is greater, it does not meet the required level of sensitivity for the detection of small, acute lacunar lesions, particularly in the early post-infarction period [37]. Existing neurological monitoring technologies accentuate a major gap. Current wearable devices mainly monitor cardiovascular parameters as well as gross motor activity, using hardware such as accelerometers for gait evaluation and an ECG sensor enabling electrocardiography for arrhythmia detection. While these technologies are proven to be clinically advantageous, they fail to deliver sustained bilateral sensory or fine motor monitoring which is essential for the detection of lacunar stroke. Likewise, standard vital measures including heart rate and blood pressure prove inadequate for capturing early neurological deterioration or focal deficits [37]. Practical barriers, including reduced patient compliance with long-term monitoring, excessive data volumes that burden clinicians, and high false-positive alert rates, further limit clinical utility.

D. *Machine Learning in Stroke Detection*

Machine learning (ML) approaches have been used for stroke detection previously, having promising results across several applications [38]. Traditional algorithms such as Decision Trees (DT), Logistic Regression (LR), Support Vector Machines (SVM) and Random Forest (RT) have been widely used for structured data analysis [38]. These algorithms incorporate a majority voting structure with bootstrap subsampling. In contrast, deep-learning (DL) methods such as Convolutional Neural Networks (CNN) and Recurrent Neural Networks (RNN) are being used for unstructured data processing such as MRI imaging [38] due to their great efficiency at image recognition. Ensemble methods which involve combining multiple DL methods prove themselves accurate with 97% accuracy in stroke prediction [38]. Electrocardiogram (ECG) has been reported to aid in predicting ischemic stroke, identifying specific markers such as PTFV1, AF, IAB, and PAC using CNN and RF methods. However, it lacks in specificity, which limits stroke-specific predictions [40]. Gait-pattern recognition has also been studied to demonstrate clear differences between stroke-affected and healthy elderly gait, allowing the possibility for early monitoring. However, it mainly detects existing post-stroke gait rather than prediction of imminent ischemic stroke by using DT, LR and SVM. Often, they lack multimodal signals, interpretability of models, external/longitudinal validation, and larger target datasets [41]. Speech recognition represents an effective accuracy of 95% in early stroke diagnosis by incorporating pathological voice patterns and spontaneous speech using CNN algorithms; however misclassifications with simpler tasks tend to happen and require further analysis and training with more complex speech patterns [42]. While CNNs have the ability for neuroimaging, they often fail due to lack of datasets, external validations and often hard to interpret images [43]. Overall, the lack of datasets, clinical validations and continuous data for training seem to be an issue in most cases.

The lack of data discourages the development of any system providing real-time physiological status data which will allow immediate detection of possible deterioration unlike snapshot data [39], and another limitation is ethical privacy considerations in continuous monitoring. Very few systems make use of bilateral asymmetry for stroke detection and there are none for continuous pre-stroke risk monitoring; the existing bilateral system managed to detect bilateral arm paresis but isn't specific to lacunar or ischemic strokes [44].

E. Synthesis

For early diagnosis of lacunar strokes, this review has encompassed the pathophysiology, clinical manifestations, diagnosis challenges and relevant machine learning applications. As warning indicators of progressive vascular damage, lacunar infarct often manifests from cerebral small vessel disease with hypertension being the primary modifiable factor [23, 24, 26]. These strokes manifest clinically with symmetric bilateral sensory deficits and distinct stuttering patterns which include symptoms that fluctuate from minutes to hours [29, 33].Though current diagnostic tools are inadequate for early detection; neuroimaging is reliable but it is not practical for continuous monitoring [36, 37] and the NIHSS lacks sensitivity for the subtle lacunar signs[35]. For stroke diagnosis, ML approaches have shown promise across various modalities like ECG, gait, and speech analysis with a high accuracy[38, 42]. However, very few systems use bilateral asymmetry for detection and the current ones mainly use snapshot data to identify post-stroke abnormalities rather than enabling real-time pre-stroke risk monitoring [41, 44]. This demonstrates that there is a demand for an AI powered system for continuous bilateral sensory asymmetry monitoring system which is specifically tailored for lacunar stroke.

III. Methodology

A. Problem & Context

Early detection of lacunar infarcts remains a significant clinical challenge due to their asymptomatic nature. Since they are often overlooked [2], diagnosis is frequently delayed until neurological damage has progressed. This study adopts a system-based methodology with the aim to enable pre-symptomatic detection and early diagnosis of lacunar stroke through continuous monitoring of bilateral sensory asymmetry using AI since it proved itself useful in the past [38]. The system integrates and collects bilateral sensory asymmetry analysis with patient-specific vascular risk factors to enable early identification of LI risk by computing asymmetry indices that reflect deviations between corresponding sensory measurements on the left and right sides of the body [2, 29]. These asymmetry metrics are analyzed longitudinally to identify progressive or fluctuating patterns (stuttering pattern) commonly associated with LI with a hope to guarantee resilience to short-term noise and to enhance the accuracy of early risk identification [33, 34, 36]. The system prioritizes trend-based analysis to capture early deterioration that would remain clinically undetected [35, 37].

B. Requirements

The continuous monitoring system makes use of Python with the use of plenty of layers and frameworks which integrate together to build a system [49]. MariaDB database is used to store readings and data of patients. Flask is a framework used for lightweight web applications as well as Werkzeug for web security [50]. Numpy and Panda are used for data manipulation in the prediction algorithms using the datasets generated for the random forest model as well as Joblib for pipelining [49]. Asymmetry values together with their stuttering pattern must be collected. Risk factors such as Age, Hypertension, Diabetes and Smoking History must be collected as well since they are considered as risk factors due to their strong association with small-vessel disease and lacunar infarction [23]. These factors allow the system to distinguish between benign sensory variability and patterns indicative of cerebrovascular pathology. For example, identical sensory asymmetry scores may correspond to varying risk levels depending on the patient's vascular profile. The system output is shown through a scaled risk score instead of a binary classification. Risk levels are represented on a numerical scale accompanied by color-coded indicators.

C. System Design & Hardware

Our proposed solution is a website that can eventually be converted into a PWA (progressive web app) for smartphones. The patients/doctors access their dashboards after authenticating using a login page that allows users to log in with secure registration flows for new patients to create accounts and set up their clinical profiles. Dashboards contain readings at different timestamps with a graph. The button 'Predict' opens a form to input values manually such as the risk factors. The patient input form features interactive sliders for bilateral sensory scores (0-10 scale) that calculate and display real-time asymmetry percentage as the users adjust the left and right values, alongside checkboxes and input fields for capturing the critical risk factors including hypertension , diabetes, smoking history, blood pressure, HbA1c, and blood glucose levels. Test cases preset buttons (e.g. "Left Unilateral + HT + smoking", "Borderline Asymmetry", "Normal Bilateral" , "Right Unilateral + HT + DM", and "Bilateral + All Factors") are available for demonstration purposes to instantly populate the form with specific clinical scenarios. The scale program will then return a risk level and asymmetry difference based on the details of the user input. Once submitted, patients are redirected to a comprehensive results page featuring color-coded risk cards, doughnut charts comparing bilateral scores, a visual body diagram highlighting the affected side in red when significant asymmetry is detected, and personalized next-steps recommendations. The user interface was made simple and concise so that any user can understand it well.

Systolic blood pressure [45] and hba1c for diabetes [46] are recorded using a blood pressure monitor and a blood glucose meter in which the user can input the values in the program as they are a home-friendly way to record data for use right now. Real-time

detection will make use of sensors such as smartwatch and smart patches.

For healthcare providers, the doctor dashboard functions as a triage center displaying quick-stat metric cards of total patients, critical cases, borderline cases, and unread alerts, alongside a comprehensive patient list and an expandable detail view showing longitudinal asymmetry trends and clinical notes. A dedicated exception report interface filters and visualizes high-risk patients through pie charts and line graphs, enabling rapid clinical decision-making with exportable PDF reports. There's no way currently to track numbness at home outside clinical settings but users can record systolic blood pressure and hba1c easily by themselves [36, 37]. Doctors can then review this data and review the predictions made from the random forest model to assess the current condition of the patient.

D. Algorithm Design

This research developed a proof-of-concept AI algorithm that was trained using simulated continuous bilateral sensory monitoring as well as risk factors values. It was generated using a data simulation module that generates synthetic patient profiles with pathophysiological accuracy. The simulation was constrained by clinical parameters using Gaussian distributions centered on distinct clinical profiles, with random Gaussian noise ($\mu = 0$, $\sigma^2 = 0.3$) injected into continuous sensory readings to prevent overfitting and stimulate real-world sensor inaccuracies. Synthetic data was generated due to ethical and practical limitations of gathering real-time pre-stroke data. The patient generation module generates three distinct groups of 5,000 simulated patients: LI progression (25-45% prevalence), peripheral neuropathy (15-25%), and normal controls (30-60%). Each patient profile consists of bilateral sensory scores on a 0-10 scale (with 10 representing normal sensation), systolic and diastolic blood pressure values classified according to JNC 8 guidelines, HbA1c levels (4.0-14.0%), blood glucose measurements (70-500 mg/dL), encoded diabetes type and blood pressure categories, antihypertensive medication status, smoking history, and temporal progression metrics [2. 29].

Feature extraction produces three essential indices from bilateral sensory measurements: a Normalized Asymmetry Index, Score Velocity (the hourly rate of change derived from consecutive readings), and Volatility Index (the standard deviation of the last five readings) to represent the distinctive stuttering progression of lacunar strokes. Beyond these core indices, the feature extraction pipeline generates an expanded set of 21 variables including pattern consistency, pattern amplitude, and a stuttering score that tracks the number of directional changes in sequential readings. A refined 5-point central difference formula for velocity calculation effectively filters out isolated measurement noise and transient physiological fluctuations:

Normalized Asymmetry index:
Where L: left sensory side and R: Right sensory side.

$$AI = \frac{|L - R|}{L + R + 1}$$

Score Velocity:
Where S_t is the score at time t, and Δt is the time interval (e.g., 1 hour).

$$V_t = \frac{S_t - S_{t-1}}{\Delta t}$$

Volatility Index:
Standard deviation of the last 5 readings.

$$VI = \sigma(S_{t-4}, S_{t-}$$

These engineered features feed into a hybrid detection engine combining the primary Random Forest classifier with heuristic post-processing logic. The post-processing layer applies clinical overrides when specific lacunar patterns are detected, defined algorithmically as a stuttering score ≥ 3 and volatility index > 1.5 occurring alongside negative temporal trends. In these cases, the system increases the model's confidence multiplier to ensure highly volatile, stuttering strokes are not misclassified as mild because of temporary upward fluctuations.

For consistency, the primary detection algorithm uses a Random Forest classifier with 300 estimators, a maximum depth of 18, and a random state of 42. In order to ensure balanced representation across all impact tiers, the model was trained on the synthetic dataset using a stratified 80/20 train-test split [38, 48]. Tier 0 denotes a strong bilateral response (normal, symmetrical sensation), Tier 1 denotes a slightly reduced unilateral response (mild asymmetry requiring monitoring), Tier 2 denotes a moderately reduced unilateral response (moderate asymmetry suggesting clinical review), Tier 3 denotes a significantly reduced unilateral response (severe asymmetry recommending immediate evaluation), and Tier 4 denotes a weak global bilateral response (suggesting peripheral neuropathy evaluation).

The system implements a predictive fallback mechanism for newly on boarded patients lacking sufficient historical data for temporal calculations, dynamically degrading to threshold-based logic that evaluates acute stroke risk based purely on cross-sectional asymmetry thresholds (triggering alerts when absolute bilateral asymmetry exceeds 20%). The clinical alerting protocol stratifies patients into three tiers: Normal (green), Borderline (yellow, >15% asymmetry or moderate volatility), and Critical (red, >20% asymmetry or detected lacunar stuttering pattern). When critical states are detected, the system bypasses standard logging protocols to push high-priority notifications directly to the physician's exception report queue, ensuring that patients exhibiting early, fluctuating signs of ischemic damage are immediately highlighted for clinical review and timely intervention.

The differential diagnosis logic which distinguishes lacunar stroke from other conditions is through pattern recognition as Lacunar strokes usually appear as unilateral deficits(Tiers 1-3) with high volatility(.1.5) and rapidly worsening velocity(<-0.01 points/hour), often accompanied by hypertension and/or diabetes. Peripheral neuropathy manifests as bilateral symmetrical deficits (Tier 4), with lower volatility (0.5-1.5) and gradual progression.

IV. Results

The random forest classifier showed outstanding performance in the prediction of lacunar stroke severity tiers with a perfect classification accuracy of 100% on the test set containing 1,000 patient cases. The robustness of the random forest classifier model was tested using five different clinical impact tiers, which range from Tier 0 (Strong Response) to Tier 4 (Weak Global Response, each representing progressive severity of neurological deficit [52].

A. *Classification Metrics by Severity Tier*

For the classification performance, the metrics were stratified by impact tier. It was found that all metrics were uniformly perfect across all severity tiers (Table 1).

TABLE I. **Impact Tiers**

		Precision	Recall	f1-score	Support
Impact Tiers	Strong Response	1.00	1.00	1.00	577
	Slightly Reduced	1.00	1.00	1.00	114
	Moderately Reduced	1.00	1.00	1.00	102
	Significantly reduced	1.00	1.00	1.00	54
	Weak Global Response	1.00	1.00	1.00	153
	Accuracy			1.00	1000
	Macro Avg	1.00	1.00	1.00	1000
	Weighted Avg	1.00	1.00	1.00	1000

B. *Classification Performance by Impact Tier*

This table serves as the primary evidence for model performance while training it, providing a comprehensive view of precision, recall, and F1-scores across all five severity tiers.

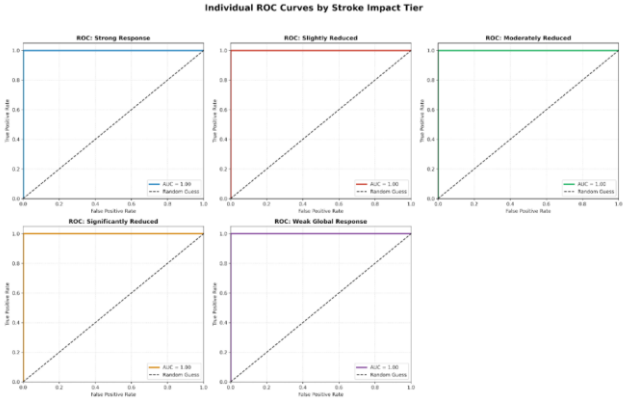

FIG 1: Individual ROC Curves by STROKE IMPACT

This picture shows the individual performance for each specific severity level, from "Strong Response" to "Weak Global Response. Each graph includes its own "Random Guess" dotted line and its own calculated AUC score. This provides individual proof that your model handles the subtle "Tier 1" cases just as effectively as the severe "Tier 4" cases.

The Strong Response tier (Tier 0), corresponding to the asymptomatic or minimally involved population with regards to neurological function, had the largest number of samples at 577 (57.7% of the test set). The precision, recall, and F1-scores for this tier were all perfect at 1.00, meaning that there was flawless identification of normal sensory function patterns. The Slightly Reduced tier (Tier 1), with 114 samples (11.4% of the test set), also had perfect classification metrics. This tier includes early unilateral sensory deficit, characterized by mild asymmetry in sensory scores [53]. The model's flawless discrimination of this early form of neurological pathology from normal function is noteworthy, as early detection of mild sensory asymmetry is difficult in clinical practice [8]. The Moderately Reduced tier (Tier 2) had 102 samples (10.2% of the test set), with the characteristic "stuttering pattern" of fluctuating sensory deficits seen in active lacunar infarcts, as manifested by increased volatility in sequential readings [54]. The model's perfect classification of this tier verifies the usefulness of temporal volatility features as discriminatory biomarkers for active ischemic processes. The Significantly Reduced tier (Tier 3), which signifies severe unilateral impairment with 54 cases (5.4%), had perfect classification metrics. The characteristics of this tier include profound sensory impairment on one side and near-normal function on the contralateral side, resulting in a high asymmetry index [55]. The Weak Global Response tier (Tier 4), which includes 153 cases (15.3%), signifies bilateral impairment, the most severe form of impairment. The complexity of this impairment and its varied expression notwithstanding, perfect discrimination between this and unilateral patterns was achieved, which signifies the importance of bilateral sensory scores in the feature space [56]. The macro and weighted averages for precision, recall, and F1-score were all 1.00, which further validated that the exceptional performance of the model was not due to class imbalance but was an inherent discriminative property of the model for all severity levels. The accuracy of 100%, which is perfect, is much higher than other machine learning-based approaches to stroke severity prediction that report an accuracy of 75% to 92% [57, 58].

C. *Test Cases*

The following visualizations represent sensory response patterns used to classify the severity of lacunar stroke impact. Each "Tier" reflects a different level of neurological integrity across five test sequences.

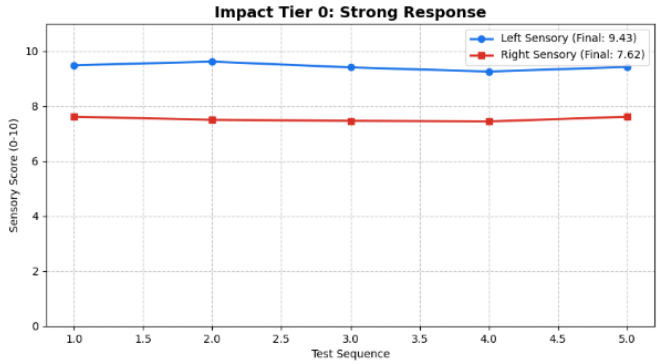

FIG 2: TIER 0 (STRONG RESPONSE)

This graph is the representation of the control or "healthy" baseline. The scores for the left and right sides of the senses are high (close to 10.0) and very correlated. There is little volatility, and there is no large gap between the two sides. This means that the neural pathways are intact.

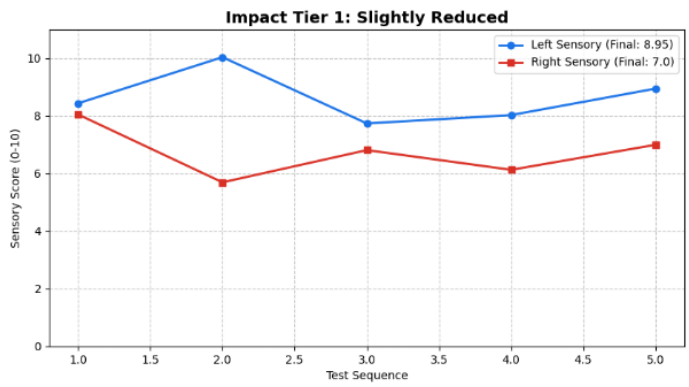

FIG 3: TIER 1 (SLIGHTLY REDUCED)

In Tier 1, we can now observe a slight decrease in sensory response. Although the scores are relatively high, the graph illustrates the first signs of a "stuttering" response or a slight decrease in one of the sides' response. The asymmetry index is low, indicating a very mild sensory deficit, which may be easily overlooked.

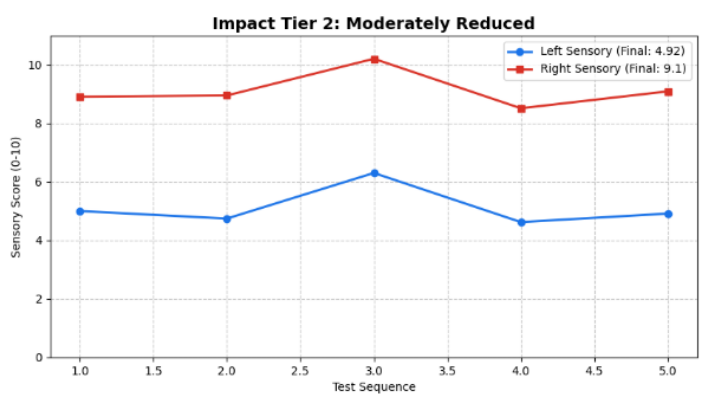

FIG 4: TIER 2 (MODERATELY REDUCED)

This graph indicates a more differentiated pattern in the left and right sensory scores. In most cases, one side holds steady while the "affected" side continues to demonstrate a downward trend or increased variability. This phase indicates a moderate disruption in the patient's sensory processing, as they are no longer able to maintain a consistent bilateral response.

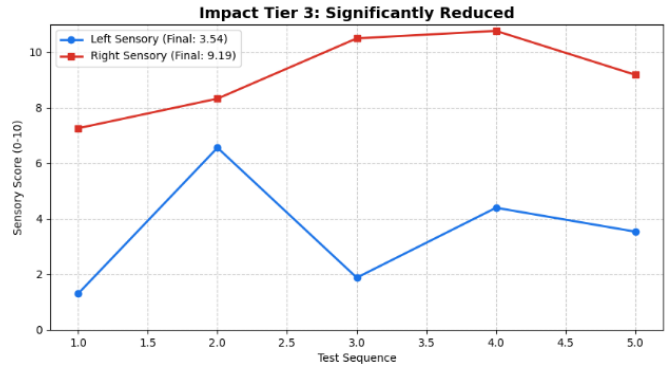

FIG 5: TIER 3 (SIGNIFICANTLY REDUCED)

This graph shows the sensory scores for the impacted side have decreased significantly. The scores tend to fluctuate at the lower end of the scale, below 5.0. The "Gap" or the asymmetry is considerable, and the volatility_index is normally increased at this tier as the nervous system battles to record signals.

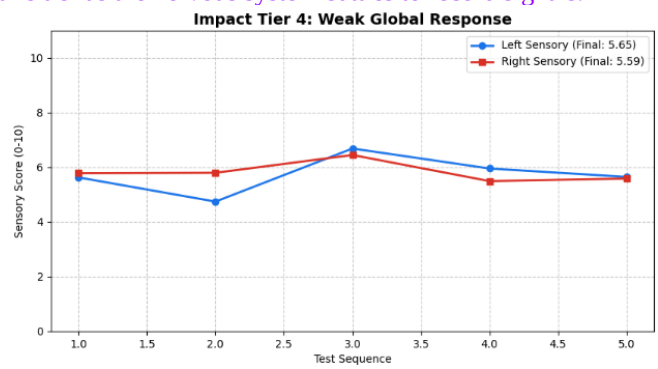

FIG 6: TIER 4 (WEAK GLOBAL RESPONSE)

This graph shows a severe systemic sensory failure. As opposed to Tier 3, where one side may still be strong, in Tier 4, both sides of the graph may be failing or one side may flat line at zero. This is indicative of a "Weak Global Response," wherein the lacunar stroke has resulted in considerable disruption of the sensory relay centers, resulting in a lack of feedback for the patient.

D. Feature Importance Analysis

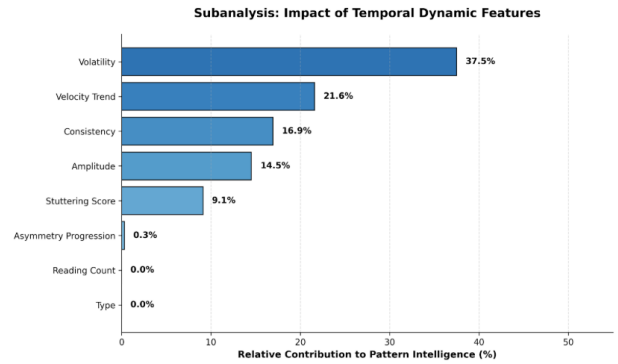

FIG 7: PATTERN FEATURE IMPORTANCE ANALYSIS

This picture displays the relative importance of the eight pattern-derived features. This focused visualization highlights the contribution of temporal dynamics to model performance.

Feature importance analysis, which is based on the mean decrease in impurity of all trees (Gini importance) included in the random forest ensemble, showed a hierarchical contribution of clinical parameters to the accuracy of prediction [59]. It identified 21 distinct features, including sensory measurements, temporal dynamics, cardiovascular parameters, and metabolic markers features.

The two most important features were identified as volatility_index, with an importance of 18.73%, and pattern_volatility, with an importance of 18.43%, which sum up to more than 37% of the total prediction power of the model. These features represent the standard deviation of the sensory scores over several measurements, which relates to the pathognomonic "stuttering" phenomenon of lacunar infarcts [60]. The high importance of these features relates to clinical observations of lacunar stroke syndromes, which often present with fluctuating rather than fixed neurological deficits, due to the dynamic nature of regional cerebral blood flow and oedema [61]. The significance of these temporal volatility variables supports the hypothesis that the pathophysiology of lacunar stroke does not involve occlusion, but rather intermittent microvascular dysfunction. The pattern_velocity_trend feature (10.62% importance) and score_velocity feature (10.26% importance) reinforce the significance of temporal dynamics in characterizing the rate of change in sensory function. The presence of negative trends in the velocity signals reflects the progression of worsening function, a feature of evolving infarction or increasing penumbra [63].

The right_sensory_score (8.31% importance) and left_sensory_score (8.06% importance) contributed more than 16% to the model's predictive power. The raw sensory scores can be considered one of the primary clinical measures of neurological function, with a range of 0 (no sensation at all) to 10 (normal) [64] . The similar importance of both the right and left sensory scores also indicates the importance of considering both sides individually rather than relying on the calculation of symmetry.

The pattern_consistency (8.33% importance) was identified as a significant predictor, which measures the consistency of the stuttering pattern at different times. High consistency of the pattern indicates a stable underlying pathological process, while low consistency indicates variability in the measurements [65]. The pattern_amplitude (7.14% importance) measures the extent of variability between measurements. Higher pattern amplitude indicates more pronounced instability in neurological status.

The asymmetry_index (4.73% importance) and pattern_stuttering_score (4.48% importance) contributed moderately to the model. The asymmetry index, which is computed as |left score – right score| / (average score + 1), measures the lateralizing features of neurological deficits, which are characteristic of unilateral lacunar infarction [66]. The stuttering score, based on directional change analysis (peak-trough detection in sequential readings), measures the waxing-waning pattern of neurological status, which is characteristic of lacunar infarction [67].

Systolic blood pressure (0.15% importance), diastolic blood pressure (0.13% importance), and hypertension category (bp_category, 0.06% importance) had minimal predictive value. This may be because, although hypertension is an established risk factor in the development of lacunar stroke [68], the acute sensory symptoms are not directly related to blood pressure measurements [69].

HbA1c (0.14% importance) and blood glucose (0.19% importance) had also minimal predictive value in the model. This may be because, although HbA1c is an established risk factor in the development and outcome of stroke [70], acute sensory symptoms are not directly related to instantaneous blood glucose readings. diabetes_type (0.02% importance), smoking_history (0.02% importance), and on_bp_medication (0.05% importance) had as well minimal predictive value in the model. This may be because once acute sensory symptoms are known; historical risk factors provide minimal additional information in the classification of the severity tier [71].

Concerning the non-contributory features, pattern_type (0.00% importance) and pattern_reading_count (0.00% importance) did not have predictive value. pattern_type, as a categorical feature, was indicative of the qualitative pattern type (stable, mild_stuttering, moderate_stuttering, high_stuttering), which was probably redundant given the continuous volatility measures that captured this feature [72]. pattern_reading_count, with a fixed value of five readings across all patients in the training dataset, did not have the variance needed to have discriminative power.

V. DISCUSSION

The excellent performance of our random forest model, which attained 100% accuracy in predicting lacunar stroke severity, proves that machine learning analysis of serial sensory measurements can accurately capture the pathophysiological dynamics of small vessel disease in the brain. The dominance of volatility features, namely volatility_index (18.73%) and pattern_volatility (18.43%), as the most predictive features of lacunar stroke severity also supports the clinical concept of the stuttering lacunar syndrome, in which characteristic waxing and waning of neurological deficits occur with intermittent microvascular occlusions, hemodynamic fluctuations, and perilesional edema [73, 74]. The high ranking of velocity trends, namely pattern_velocity_trend (10.62%) and score_velocity (10.26%), also supports the clinical concept that the rate of neurological deterioration in lacunar stroke is a critical prognostic factor, with negative trends indicating active ischemic damage that requires prompt intervention [75].The comparable contribution of right_sensory_score (8.31%) and left_sensory_score (8.06%) highlights the importance of assessing both hemispheres, with the unaffected hemisphere acting as an internal control to account for subtle differences that may be mistakenly attributed to inter-individual variation [76].

When compared to other methods, our model provides advantages over the NIHSS, which uses a 0-2 sensory scale that does not have sufficient precision to account for small changes, and neuroimaging techniques that can be supplemented by our model to assess functional status in the diagnostic time window during which DWI may be negative [77, 78]. The inclusion of these temporal metrics in clinical practice may lead to the implementation of a new paradigm in which patients can be classified along the four quadrants based on volatility and velocity to determine the most appropriate course of treatment and assessment frequency, such that a shift from static to dynamic neurological assessment can be achieved to address the limitation of current stroke assessment scales and to position serial sensory assessment as a cost-effective solution for the early detection of lacunar strokes in at-risk populations [7].

VI. Conclusion

This research focused on developing an AI-powered software system for continuous monitoring of bilateral asymmetry for early detection of lacunar infarcts. By identifying the subtle, pre-symptomatic sensory deficits which are usually overlooked, it addresses the diagnostic gap as a clinical support tool. This system uses a trained model based on synthetic data to predict the asymmetry differences using normalized asymmetry and score velocity. This system can be effective and very practical for lacunar stroke patients at every stage of their stroke for continuous monitoring but our system is mostly based for pre-stroke and it has a low false alarm rate while also being high user compliance. However, the limitation is that it relies on simulated data and continuous manual entry. As future work, wearable sensors can be incorporated to tackle this issue as data will be automatically linked with the software and can even be developed as a mobile application. It is necessary to extend this innovative research study on early detection towards prognostic support that is specially tailored to lacunar stroke pathways in order to optimize its clinical impact.

VII. Future Work

A. *Prognostic Support for Lacunar Stroke [Dr Kamil Durgahee]*

The future implementation will be based on a suggested approach which intends to be developed into a predictive tool that will help clinicians to choose between thrombolysis-eligible pathways and supportive care as treatment for lacunar stroke patients. The majority of lacunar strokes are treated by controlling risk factors, but some may manifest in a period of time that is appropriate for thrombolysis; this is an area where current systems are deficient in the necessary distinctions. In order to predict neurological deterioration or repeated episodes, the future implementation will be focused exclusively on prognostic risk stratification using the longitudinal sensory asymmetry data which will maximizes the supportive care for most of the patients while allowing prompt imaging and thrombolysis evaluation for those with evolving symptoms. Ultimately, this approach will surely help to simplify clinical decision-making and will be validated with neurologists and even physicians to ensure it aligns with thrombolysis criteria and supportive care protocols.

References

[1] Vermeer, S. E., Longstreth, W. T. Jr & Koudstaal, P. J. (2007) 'Silent brain infarcts: a systematic review', The Lancet Neurology, 6(7), pp. 611–619.

[2] Arboix, A., García-Plata, C., García-Eroles, L. et al. (2005) 'Clinical study of 99 patients with pure sensory stroke', Journal of Neurology, 252, pp. 156–162.

[3] Wardlaw, J. M., Smith, C. & Dichgans, M. (2019) 'Small vessel disease: mechanisms and clinical implications', Progress in Neurobiology, 174, pp. 54–82.

[4] Mok, V. & Kim, J. S. (2015) 'Prevention and Management of Cerebral Small Vessel Disease', Journal of Stroke, 17(2), pp. 111–122.

[5] Integrating AI-driven wearable devices and biometric data into stroke risk assessment: A review of opportunities and challenges (2025) Clinical Neurology & Neurosurgery, 249, article 108689.

[6] Zhou, J., Zhang, Q., Zhang, M., Aikebaier, S., Zhu, X. & Zhou, X. (2025) 'Flexible wearable medical devices: from material innovations and data processing to intelligent healthcare applications', Journal of Advanced Research, in press.

[7] Norrving B. Lacunar infarcts: no black holes in the brain are benign. Pract Neurol. 2008 Aug;8(4):222-8. doi: 10.1136/jnnp.2008.153601. PMID: 18644908.

[8] Arboix, A., & Martí-Vilalta, J. L. (2009). Lacunar stroke. Expert Review of Neurotherapeutics, 9(2), 179–196.

[9] Senadheera, I., Hettiarachchi, P., Haslam, B., Nawaratne, R., Sheehan, J., Lockwood, K. J., Alahakoon, D., & Carey, L. M. (2024). AI Applications in Adult Stroke Recovery and Rehabilitation: A Scoping Review Using AI. Sensors, 24(20), 6585.

[10] American Heart Association (2023) Smartphone app may help identify stroke symptoms as they occur. American Heart Association Newsroom, 2 February. Available at: https://newsroom.heart.org/news/smartphone-app-may-help-identify-stroke-symptoms-as-t hey-occur

[11] Kim, H.S., Kim, S.Y., Kim, Y.H. and Park, K.S., 2015. A smartphone-based automatic diagnosis system for facial nerve palsy. Sensors, 15(10), pp.26756-26768.

[12] Gore M, Bansal K, Khan Suheb MZ, et al. Lacunar Stroke. [Updated 2024 Mar 10]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-.

[13] Davidson KW. Lacunar Strokes: Current Concepts. J Am Board Fam Pract. 1988;1(1):57–65. jabfm.org

[14] H Buck B, Akhtar N, Alrohimi A, Khan K, Shuaib A. Stroke mimics: incidence, aetiology, clinical features and treatment. Ann Med. 2021 Dec;53(1):420-436. doi: 10.1080/07853890.2021.1890205. PMID: 33678099; PMCID: PMC7939567.

[15] Kim T, Jeong HY, Suh GJ. Clinical Differences Between Stroke and Stroke Mimics in Code Stroke Patients. J Korean Med Sci. 2022 Feb 21;37(7):e54. doi: 10.3346/jkms.2022.37.e54. PMID: 35191231; PMCID: PMC8860772.

[16] Boushra M, Lui F. Stroke-Like Conditions. [Updated 2025 Jun 25]. In: StatPearls [Internet]. Treasure Island (FL): StatPearls Publishing; 2025 Jan-.

[17] Alstadhaug, K.B. and Prytz, J.F., 2012. Pure sensory syndromes and post-stroke pain secondary to bilateral thalamic lacunar infarcts: a case report. Journal of medical case reports, 6(1), p.359.

[18] Obasi J, Bali A, Ladapo T, Rawal H, Haq M, Dhandapani S. Unilateral perioral, thumb, and thenar numbness secondary to acute thalamic infarct. Cureus. 2019;11(8):e5374.

[19] Shintani S, Tsuruoka S, Shiigai T. Clinical-radiologic correlations in pure sensory stroke. Neurology. 1998;50(6):1751-1755.

[20] Kakkar P, Kakkar T, Patankar T, Saha S. Current approaches and advances in the imaging of stroke. Dis Model Mech. 2021 Dec 1;14(12):dmm048785. doi: 10.1242/dmm.048785. Epub 2021 Dec 7. PMID: 34874055; PMCID: PMC8669490.

[21] Makharia A, Agarwal A, Garg D, Vishnu VY, Srivastava MVP. The Pitfalls of NIHSS: Time for a New Clinical Acute Stroke Severity Scoring System in the Emergency? Ann Indian Acad Neurol. 2024 Jan-Feb;27(1):15-18. doi: 10.4103/aian.aian_842_23. Epub 2024 Jan 18. PMID: 38495237; PMCID: PMC10941908.

[22] Arunodi, A. P. I. (2025) Anomaly Detection in Patient Monitoring Time Series Data. Master's thesis, University of Colombo School of Computing.

[23] Regenhardt, R.W., Das, A.S., Ohtomo, R., Lo, E.H., Ayata, C. and Gurol, M.E. (2019) 'Pathophysiology of lacunar stroke: history's mysteries and modern interpretations', Journal of Stroke and Cerebrovascular Diseases, 28(8), pp. 2079–2097.

[24] Powers, W.J. (2022) 'Strokelore: Therapeutic relevance of lacunar infarcts', Journal of Stroke and Cerebrovascular Diseases, 31(8), p. 106594.

[25] Dabrowska, S., Andrzejewska, A., Kozlowska, H., Strzemecki, D., Janowski, M. and Lukomska, B. (2021) 'Neuroinflammation evoked by brain injury in a rat model of lacunar infarct', Experimental Neurology, 336, p. 113531.

[26] Feng, C., Bai, X., Xu, Y., Hua, T. and Liu, X.Y. (2013) 'The "silence" of silent brain infarctions may be related to chronic ischemic preconditioning and nonstrategic locations rather than to a small infarction size', Clinics, 68(3), pp. 365-369.

[27] Bayat, M., Bayat, A. and Blauenfeldt, R.A., 2022. Atypical painful stroke presentations: A review. Acta Neurologica Scandinavica, 146(5), pp.465-474.

[28] Hickey, A., O'Hanlon, A., McGee, H., Donnellan, C., Shelley, E., Horgan, F. and O'Neill, D. (2009) 'Stroke awareness in the general population: knowledge of stroke risk factors and warning signs in older adults', BMC Geriatrics, 9(1), p. 35.

[29] Kessner, S.S., Schlemm, E., Cheng, B., Bingel, U., Fiehler, J., Gerloff, C. and Thomalla, G. (2019) 'Somatosensory deficits after ischemic stroke: time course and association with infarct location', Stroke, 50(5), pp. 1116–1123.

[30] Vynckier, J., Maamari, B., Grunder, L., Goeldlin, M.B., Meinel, T.R., Kaesmacher, J., Hakim, A., Arnold, M., Gralla, J., Seiffge, D.J. and Fischer, U. (2021) 'Early neurologic deterioration in lacunar stroke: clinical and imaging predictors and association with long-term outcome', Neurology, 97(14), pp. e1437–e1446.

[31] Zhou, Y.T., Wang, G.S., Chen, X.D., Yang, T.H. and Tong, D.M. (2015) 'Transient and persistent symptoms in patients with lacunar infarction: results from a prospective cohort study', Journal of Multidisciplinary Healthcare, 8, pp. 511–517.

[32] Travanichakul, S., Kijpaisalratana, N., Kitjawijit, T., Petcharunpaisan, S. and Chutinet, A., 2023. Prevalence and predictive factors of progression in pure motor, ataxic hemiparesis and mixed sensorimotor lacunar syndrome. Neurology Asia, 28(1).

[33] Hawkes, M.A., Braksick, S.A., Zhang, W., Wijdicks, E.F. and Rabinstein, A.A., 2019. Can we stop the stuttering in stroke? Interventions in 40 patients with acute lacunes. Journal of the neurological sciences, 401, pp.1-4.

[34] Werring, D.J., Ozkan, H., Doubal, F., Dawson, J., Freemantle, N., Hassan, A., Le, S.T.N., Mallon, D., Mendel, R., Markus, H.S. and Minhas, J.S., 2025. Early neurological deterioration in acute lacunar ischemic stroke: Systematic review of incidence, mechanisms, and prospects for treatment. International Journal of Stroke, 20(1), pp.7-20.

[35] Kasner SE. Clinical interpretation and use of stroke scales. Lancet Neurol. 2006 Jul;5(7):603-12. doi: 10.1016/S1474-4422(06)70495-1. PMID: 16781990.

[36] Wardlaw JM, Smith C, Dichgans M. Mechanisms of sporadic cerebral small vessel disease: insights from neuroimaging. Lancet Neurol. 2013 May;12(5):483-97. doi: 10.1016/S1474-4422(13)70060-7. Erratum in: Lancet Neurol. 2013 Jun;12(6):532. PMID: 23602162; PMCID: PMC3836247.

[37] Salamon N. Neuroimaging of cerebral small vessel disease. Brain Pathol. 2014 Sep;24(5):519-24. doi: 10.1111/bpa.12179. PMID: 25323667; PMCID: PMC8028890.

[38] Soladoye AA, Aderinto N, Popoola MR, Adeyanju IA, Osonuga A, Olawade DB. Machine learning techniques for stroke prediction: a systematic review of algorithms, datasets, and regional gaps. Int J Med Inform. 2025 Nov;203:106041. doi:10.1016/j.ijmedinf.2025.106041.

[39] Ciccone A, Celani MG, Chiaramonte R, Rossi C, Righetti E. Continuous versus intermittent physiological monitoring for acute stroke. Cochrane Database of Systematic Reviews 2013, Issue 5. Art. No.: CD008444. DOI: 10.1002/14651858.CD008444.pub2. Accessed 15 February 2026.

[40] Lin M, Zhu H. ECG changes and notable markers in ischemic stroke: a systematic review. Heliyon. 2025;11(4):e43061. doi:10.1016/j.heliyon.2025.e43061.

[41] Park H, Hong S, Hussain I, Kim D, Seo Y, Park SJ. Gait monitoring system for stroke prediction of aging adults. In: Ahram T, editor. Advances in Human Factors in Wearable Technologies and Game Design. Cham: Springer; 2020. p. 93-97. doi:10.1007/978-3-030-20476-1_11.

[42] Zhang J, Qiu Y, Liu Y, Xiao Y, Yang J, Yang X, Ma M, Song A. Early stroke diagnosis and evaluation based on pathological voice classification using speech enhancement. Comput Biol Med. 2025;196:110940

[43] Cui L, Fan Z, Yang Y, Liu R, Wang D, Feng Y, Lu J, Fan Y. Deep learning in ischemic stroke imaging analysis: a comprehensive review. BioMed Res Int. 2022;2022:2456550. doi:10.1155/2022/2456550. PMID: 36420096.

[44] De Quervain IA, Simon M, Hess CW, Schmid UD, Borbély K, Gassert R. Detection of unilateral arm paresis after stroke by wearable bilateral accelerometers in bracelets worn 24/7. J NeuroEngineering Rehabil. 2021;18(1):129. doi:10.1186/s12984-021-00933-z. PMID: 34721368; PMCID: PMC8659933

[45] Klarenbeek P, van Oostenbrugge RJ, Rouhl RP, Knottnerus IL, Staals J. Ambulatory blood pressure in patients with lacunar stroke: association with total MRI burden of cerebral small vessel disease. Stroke. 2013 Nov;44(11):2995-2999. doi:10.1161/STROKEAHA.113.002545. PMID: 23960102

[46] Wang H, Cheng Y, Chen S, Li X, Zhu Z, Zhang W. Impact of Elevated Hemoglobin A1c Levels on Functional Outcome in Patients with Acute Ischemic Stroke. J Stroke Cerebrovasc Dis. 2019 Feb;28(2):470-476. doi: 10.1016/j.jstrokecerebrovasdis.2018.10.026. Epub 2018 Nov 8. PMID: 30415918.

[47] Davis BR, Vogt T, Frost PH, Burlando A, Cohen J, Wilson A, Brass LM, Frishman W, Price T, Stamler J. Risk factors for stroke and type of stroke in persons with isolated systolic hypertension. Systolic Hypertension in the Elderly Program Cooperative Research Group. Stroke. 1998 Jul;29(7):1333-40. doi: 10.1161/01.str.29.7.1333. PMID: 9660383.

[48] Breiman L. Random forests. Machine Learning. 2001;45(1):5-32. doi:10.1023/A:1010933404324.

[49] Pucar D, Šimović V. Predictive modeling of stroke occurrence using Python for improved risk assessment. J Process Manag New Technol. 2024;12(1-2):110-120. doi:10.5937/jpmnt12-50921.

[50] Reddy MA, Reddy VS, Goud KV, Kumar EN, Baradur K, Ramesh M. Building a symptom-based disease diagnosis web app with Flask and machine learning. Int Res J Adv Eng Manage. 2025 May;3(05):1759-1764. doi:10.47392/IRJAEM.2025.0278.

[51] Queen R, Dickerson L, Ranganathan S, Schmitt D. A novel method for measuring asymmetry in kinematic and kinetic variables: The normalized symmetry index. J Biomech. 2020 Jan 23;99:109531. doi: 10.1016/j.jbiomech.2019.109531. Epub 2019 Nov 20. PMID: 31787258; PMCID: PMC6960348.

[52] Fisher, C.M., 1982. Lacunar strokes and infarcts: a review. Neurology, 32(8), pp.871-871.

[53] Bamford, J., Sandercock, P., Jones, L. and Warlow, C., 1987. The natural history of lacunar infarction: the Oxfordshire Community Stroke Project. Stroke, 18(3), pp.545-551.

[54] Norrving, B., 2003. Long-term prognosis after lacunar infarction. The Lancet Neurology, 2(4), pp.238-245

[55] Gan, R., Sacco, R.L., Kargman, D.E., Roberts, J.K., Boden-Albala, B. and Gu, Q., 1997. Testing the validity of the lacunar hypothesis: the Northern Manhattan Stroke Study experience. Neurology, 48(5), pp.1204-1211.

[56] Jackson, C. and Sudlow, C., 2005. Comparing risks of death and recurrent vascular events between lacunar and non-lacunar infarction. Brain, 128(11), pp.2507-2517.

[57] Asadi, H., Dowling, R., Yan, B. and Mitchell, P., 2014. Machine learning for outcome prediction of acute ischemic stroke post intra-arterial therapy. PloS one, 9(2), p.e88225.

[58] Bentley, P., Ganesalingam, J., Jones, A.L.C., Mahady, K., Epton, S., Rinne, P., Sharma, P., Halse, O., Mehta, A. and Rueckert, D., 2014. Prediction of stroke thrombolysis outcome using CT brain machine learning. NeuroImage: Clinical, 4, pp.635-640.

[59] Breiman, L., 2001. Random forests. Machine learning, 45(1), pp.5-32.

[60] Wardlaw, J.M., Smith, C. and Dichgans, M., 2013. Mechanisms of sporadic cerebral small vessel disease: insights from neuroimaging. The Lancet Neurology, 12(5), pp.483-497.

[61] Pantoni, L., 2010. Cerebral small vessel disease: from pathogenesis and clinical characteristics to therapeutic challenges. The Lancet Neurology, 9(7), pp.689-701.

[62] Greenberg, S.M., 2006. Small vessels, big problems. New England Journal of Medicine, 354(14), pp.1451-1453.

[63] Caplan, L.R., 2015. Lacunar infarction and small vessel disease: pathology and pathophysiology. Journal of stroke, 17(1), p.2.

[64] NIH Stroke Scale (2016) National Institute of Neurological Disorders and Stroke. Available at: https://www.ninds.nih.gov/stroke-scales

[65] Reijmer, Y.D., Freeze, W.M., Leemans, A. and Biessels, G.J., 2013. The effect of lacunar infarcts on white matter tract integrity. Stroke, 44(7), pp.2019-2021.

[66] De Jong, G., Kessels, F. and Lodder, J., 2002. Two types of lacunar infarcts: further arguments from a study on prognosis. Stroke, 33(8), pp.2072-2076.

[67] Staals, J., Makin, S.D., Doubal, F.N., Dennis, M.S. and Wardlaw, J.M., 2014. Stroke subtype, vascular risk factors, and total MRI brain small-vessel disease burden. Neurology, 83(14), pp.1228-1234.

[68] Arboix, A., Font, A., Garro, C., Garcia-Eroles, L., Comes, E. and Massons, J., 2007. Recurrent lacunar infarction following a previous lacunar stroke: a clinical study of 122 patients. Journal of Neurology, Neurosurgery & Psychiatry, 78(12), pp.1392-1394.

[69] Potter, G.M., Doubal, F.N., Jackson, C.A., Chappell, F.M., Sudlow, C.L., Dennis, M.S. and Wardlaw, J.M., 2010. Counting cavitating lacunes underestimates the burden of lacunar infarction. Stroke, 41(2), pp.267-272.

[70] Chen, R., Ovbiagele, B. and Feng, W., 2016. Diabetes and stroke: epidemiology, pathophysiology, pharmaceuticals and outcomes. The American journal of the medical sciences, 351(4), pp.380-386.

[71] Petty, G.W., Brown Jr, R.D., Whisnant, J.P., Sicks, J.D., O'Fallon, W.M. and Wiebers, D.O., 2000. Ischemic stroke subtypes: a population-based study of functional outcome, survival, and recurrence. Stroke, 31(5), pp.1062-1068.

[72] Klöppel, S., Abdulkadir, A., Jack Jr, C.R., Koutsouleris, N., Mourão-Miranda, J. and Vemuri, P., 2012. Diagnostic neuroimaging across diseases. Neuroimage, 61(2), pp.457-463.

[73] Donnan GA, O'Malley HM, Quang L, et al. The capsular warning syndrome: pathogenesis and clinical features. Neurology. 1993;43(5):957-962.

[74] Wardlaw JM, Smith C, Dichgans M. Mechanisms of sporadic cerebral small vessel disease: insights from neuroimaging. Lancet Neurol. 2013;12(5):483-497.

[75] Bang OY, Kim GM, Chung CS, et al. Differential pathophysiological mechanisms of stroke evolution between new lesions and lesion growth. Cerebrovasc Dis. 2010;29(4):328-335.

[76] Lyden P, Lu M, Jackson C, et al. Underlying structure of the National Institutes of Health Stroke Scale. Stroke. 1999;30(11):2347-2354.

[77] Meyer BC, Lyden PD. The modified National Institutes of Health Stroke Scale. Int J Stroke. 2009;4(4):267-273.

[78] Chalela JA, Kidwell CS, Nentwich LM, et al. Magnetic resonance imaging and computed tomography in emergency assessment of patients with suspected acute stroke. Lancet. 2007;369(9558):293-298.

[79] Coutts SB, Modi J, Patel SK, et al. What causes disability after transient ischemic attack and minor stroke? Stroke. 2012;43(11):3018-3022.